Analysis of the PheB24Ser mutation to the Human Insulin protein (2HIU) using Molecular Dynamics Simulations

Harry Wang

Introduction

About Insulin

Insulin is a very important messenger molecule in the human body. When the sugar levels in the human body are high, it gets released into the blood as a signal. It then binds to insulin receptors in the liver telling the liver to remove glucose from the blood and store it.

Insulin Mutations

There are several known mutations to insulin. One of them, known as Insulin Los Angeles, is a mutation from phenylalanine to serine at the position 24 of chain B, known as PheB24Ser. It is said that it “perturbs the “aromatic anchor” of the B-chain b-strand (B24-B28)” and decreases the foldability of the protein. In addition, it is said to reduce alpha helix content of the protein. (Dhayalan, 2022). This was proven when they took a CD spectrum to understand the structure of the protein. As seen in **Figure 1**, the SerB24 mutation had a lower absorption, meaning that the alpha helix content may have been lower. Another paper that also did the CD-spectrum also found that the SerB24 mutated protein had less activity, saying that it would inhibit “the ability to form quaternary structure” (Wollmer, 1981).

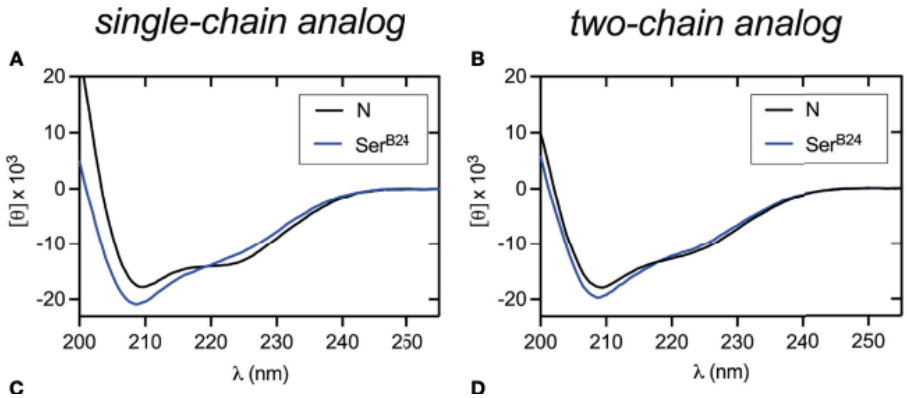


Figure 1: CD spectrums of human insulin and the SerB24 mutation

Another source says that this mutation causes hyperinsulinemia, a disease where there is too much insulin in the blood due to cells not responding to the insulin. It is likely due to reduced

activity caused by the mutation. As seen in **Figure 2**, the biological activity was significantly reduced in the SerB24, the mutated version, to just 0.6%-1.5%. (Nishi, 2011)

	Normal Human	Chicago LeuB25	Los Angeles SerB24	Wakayama LeuA3
In vitro receptor binding (%)	100	2.0–2.8	0.5–1.8	0.3–0.7
Biological activity (%)	100	1.9–2.5	0.6–1.5	0.4–0.8
In vivo serum half-life in dog (min)	3.6	15.0	24.5	35.0

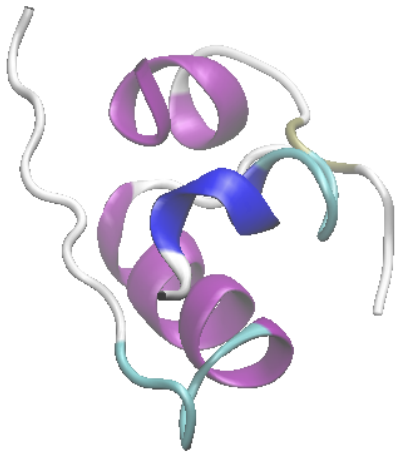
Figure 2: Reduced insulin activity under the B24 mutation (Nishi, 2011)

Another study looked at the mutation of the phenylalanine to biphenylalanine (not an amino acid) in the same position (B24). In this study, the insulin with this mutation had no activity at all. The study concluded that it means that the phenylalanine at position B24 should be “crucial for insulin activity” (Du, 2008). Thus, it agrees with the other papers.

In this paper, I will look at how the PheB24Ser mutation affects the function of human insulin.

Structure

Insulin is composed of two amino acid chains linked by disulfide bridges, with one being 21 residues long and one being 30 residues long. It has a total of 786 atoms (Hua, 1996). As seen in the diagram below, it has three alpha helices and a few turns.



Importance of MD Simulations

I chose to use Molecular Dynamics simulations because they could simulate the movement of a protein and allow for me to understand how it interacts with its surroundings. Unlike crystal structures, they can allow me to understand the movement of insulin without doing experiments that might be time consuming and expensive.

Methods

Molecular dynamics simulations were used to model the movement of the protein. Visual Molecular Dynamics was used to visualize and analyze the trajectories of the protein while a supercomputer was used to compute the simulations. The following steps were done to set up and simulate the protein and the mutation.

Setup

The protein was downloaded as a PDB file from the RCSB website containing the coordinates of every atom in the protein. The following steps were done to prepare the protein for simulations:

1. **Adding hydrogens:** Since the original PDB file did not have a high enough resolution and did not contain hydrogens, I used the autoPSF feature in VMD to add hydrogens. This generated a .psf file storing the connectivities of the atoms in the protein.
2. **Solvation:** I added a water box around the protein to simulate an authentic environment where the protein was dissolved in water. I rotated the protein to minimize the volume and padded 10 angstroms on each side of the box. This was because periodic boundary conditions were used, where a solvated protein identical to the one simulated would appear on the ends of the box. Thus, I needed to separate the protein from the other proteins. The maximum distance in which two molecules can interact in the simulation is 10 angstroms. The 10 angstrom padding, added to the 10 angstrom padding on the other box in the periodic boundary, gives a 20 angstrom distance from one protein to the other, giving it extra space.
3. **Ionization:** I added positive and negative salt molecules (Na+ and Cl-) to the protein to simulate an authentic biological environment. I set the concentration to 0.1 mol/L.
4. **Centering:** I centered the solvation box to make analysis easier in the future using TCL commands.
5. **Restraining:** I added restraints to the backbone of the protein to stabilize the protein's structure during the heating up. I set the beta value to 10 in the pdb file.

To prepare the simulation for the mutated version of my protein, I created a mutated version of the protein using the "Mutate Residue" function in VMD and stored it in a .pdb file. I then did the same steps as I did for the control protein.

Minimization

After I successfully solvated, ionized, and restrained my protein, I began to simulate the protein. I started by minimizing the energy in order to stabilize the protein and avoid clashes. I ran the minimization for a total of 10000 steps, with each step lasting one femtosecond. The potential energy between the molecules was successfully reduced. The data was written into a .dcd trajectory file, with 100 steps per frame. Hence, the .dcd file had 100 frames in total.

Heatup and Equilibration

After the minimization, I submitted a job to heat up the protein and equilibrate it with the environment. The heatup process started from 240K and ended with 300K, with a 5K increment. 10,000 steps were simulated for temperature point, for a total of 130,000 steps in total. After this was done, I simulated the protein for another 100,000 steps. After that, I gradually started removing the restraints. The specifics are as follows:

1. Constraint Scaling at 0.75 for 100,000 steps
2. Constraint Scaling at 0.5 for 100,000 steps
3. Constraint Scaling at 0.25 for 120,000 steps
4. Constraint Scaling at 0.125 for 150,000 steps
5. Constraint Scaling at 0.05 for 150,000 steps

When all constraints were removed, I simulated the protein for an additional 2,000,000 steps. In total, the simulation lasted for 2,850,000 steps, or 2.85 nanoseconds.

Production Runs

When the protein was successfully equilibrated with the environment, I began to simulate the protein moving. The simulations were done with NPT (pressure and temperature being kept constant). In total, 50 nanoseconds, or 50 million steps were simulated, divided into two 25-million-step sections in order to reduce the potential computing power use if things went wrong. All restraints were removed by this point and the protein was free to move around.

Water/Salt Removal and Concatenating

After all of the simulations were done, the .dcd trajectory files of the equilibration simulation and the production runs were concatenated into one trajectory file for analysis using a script called Catdcd. In addition, the water and salt molecules were removed from the trajectory as they weren't relevant. The combined trajectory had 52.85 million steps, equal to 52.85 nanoseconds. It had 1 frame for 2000 steps of simulation. This was further reduced by a factor of 10 when the trajectory was concatenated. Hence, the final trajectory had a total of 2645 frames.

Results

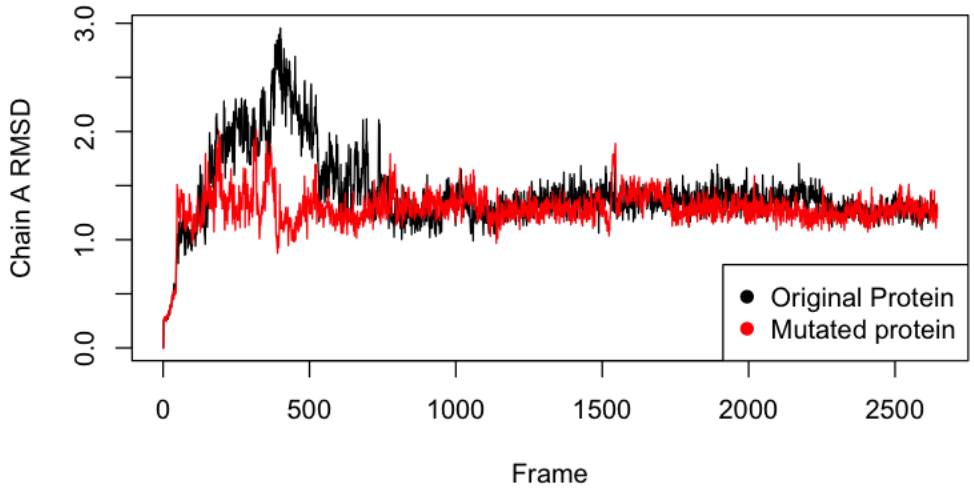
Several different variables were measured for the simulation. This included:

- **Root Mean Square Deviation (RMSD)**, the amount in which the radius of the protein deviated from the original value. It is a measure used to identify large conformational changes in the protein over time. This was taken for every frame in the simulation.
- **Root Mean Square Fluctuation (RMSF)**, the root mean square amount in which each residue of the protein deviates from a reference position. It can be used to compare the relative amounts that the residues moved by. This was taken for every residue in the protein.
- **Radius of gyration (Rg)**, a measure of the protein's radius over time. It was measured for every frame in the simulation.
- Changes in the secondary structure over time

These analyses were done for both the control protein and the mutated protein. In addition, some of the measurements, such as the RMSD and RMSF, were done separately for the two chains in the proteins.

Root Mean Square Deviation

Chain A



As seen in this graph, the RMSD trajectory of chain A of the mutated protein was quite similar to the original protein. The RMSD during the heatup was very similar, and the RMSD values converted to similar values for both proteins in the end. However, a notable difference was that between approximately frame 250 and frame 750, the control protein's chain A RMSD increased

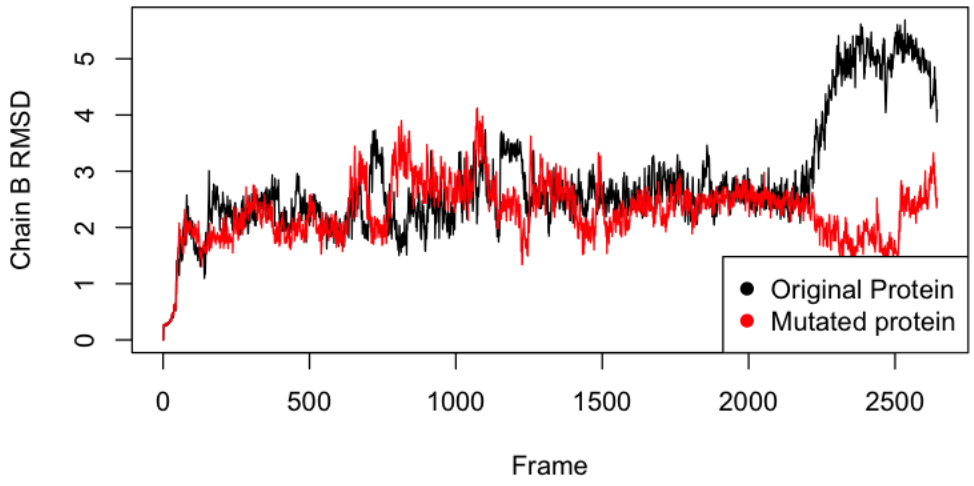
significantly, from about 1 angstrom to almost 3 angstroms and then decreased back to the original value. In comparison, the mutated protein's chain A RMSD almost did not increase during this period. To look at what caused this increase, I looked at the trajectory and found the following:

Frame 500:

Frame 1000

As seen in this picture, at frame 500, when the RMSD for chain A of the control protein peaked, it "stretched" itself and made it longer compared to before. This is likely what caused the increase in the RMSD and the subsequent decrease.

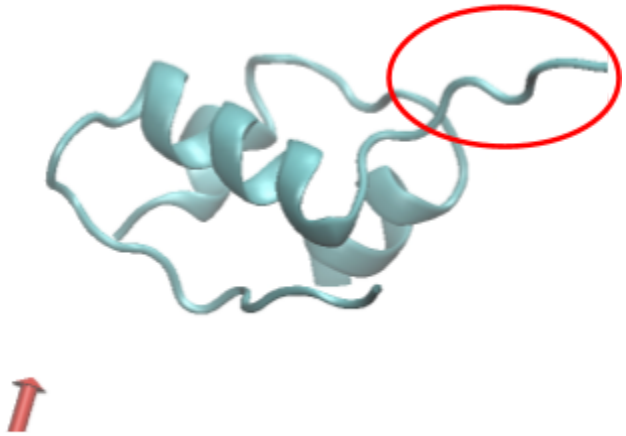
Chain B



As seen in this graph, a notable observation is that the RMSDs for chain B on both proteins are much higher than chain A on average. The average RMSD when the protein equilabilized was around 2.5 angstroms compared to about 1.25 for chain A. The RMSD trajectory of chain B of both proteins was similar until about frame 2100. After this point, the RMSD of the original protein increased dramatically, from about 3 angstroms to more than 5 angstroms, and stayed at that level until the end of the simulation, though there were signs of it decreasing at the very end. By comparison, the mutated protein's chain B RMSF barely fluctuated. In fact, it decreased a bit when the rapid increase for the control protein happened. When I looked at the trajectory of the original protein, I observed the following:

Frame 2000:

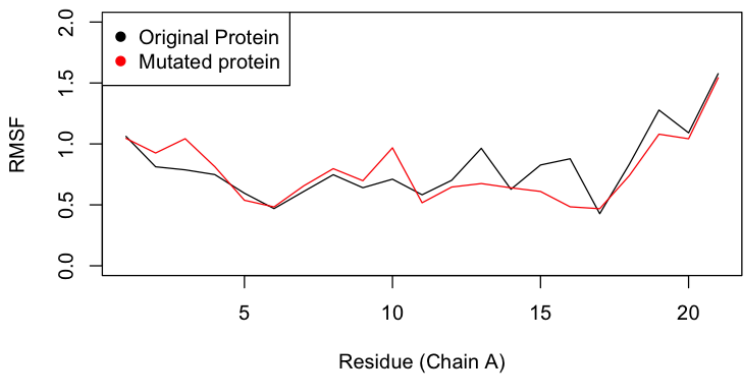
Frame 2500:



As seen in this picture, a tail on chain B ended up "unfolding" between frame 2000 and frame 2500. This is likely what caused the dramatic increase in the RMSD for chain B of the control protein.

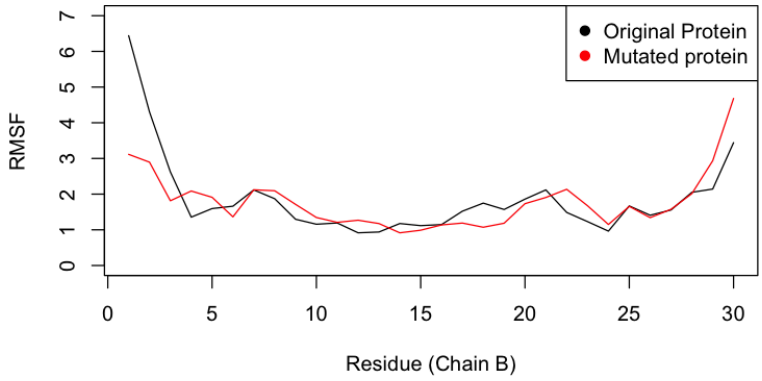
Root Mean Square Fluctuation

Chain A



As seen in this chart, the RMSF values of chain A of the two proteins were very much similar. In general, the mutated protein's chain A had higher RMSF values for the first 10 residues and lower RMSF values for the rest. However, the difference was very small.

Chain B



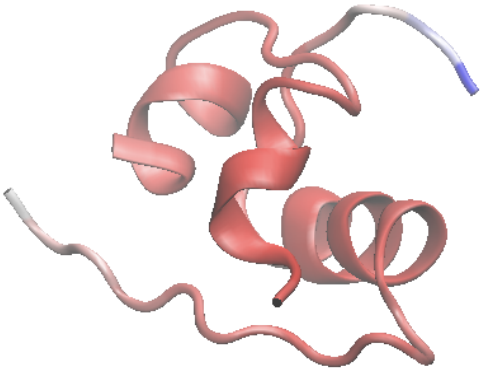
As seen in this diagram, the RMSF for chain B of both proteins was also very similar. One notable difference, however, was that residues 0, 1, and 2 of chain B of the original protein had significantly higher RMSF values than chain B of the mutated protein. Also, residue 29 and 30 of the mutated protein had slightly higher RMSF values than the original protein.

The large RMSF values of the first few residues of chain B of the control protein could be explained by the “unfolding” of a turn in the protein (see the RMSD analysis) which also caused sudden RMSD increase at the end for the control protein.

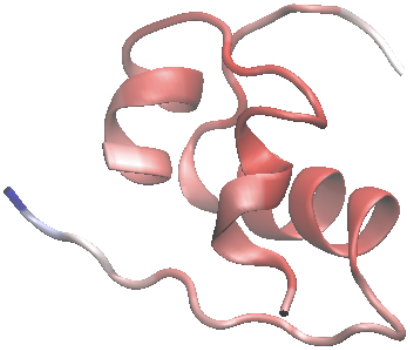
Proteins Color-coded Based on RMSF

I wrote a script to color-code parts of the protein based on their RMSF. There are shown as follows:

Control:



Mutation:

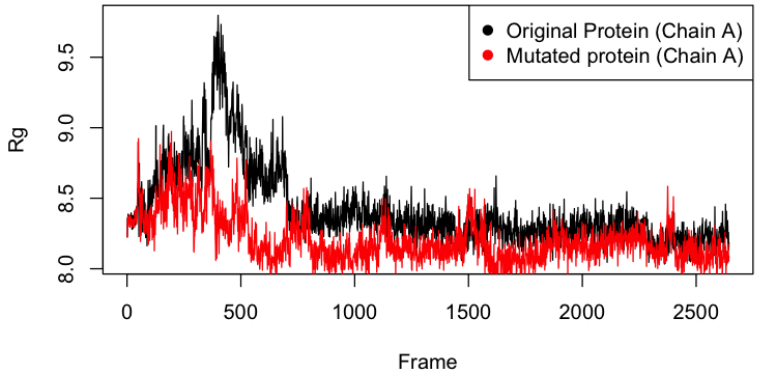


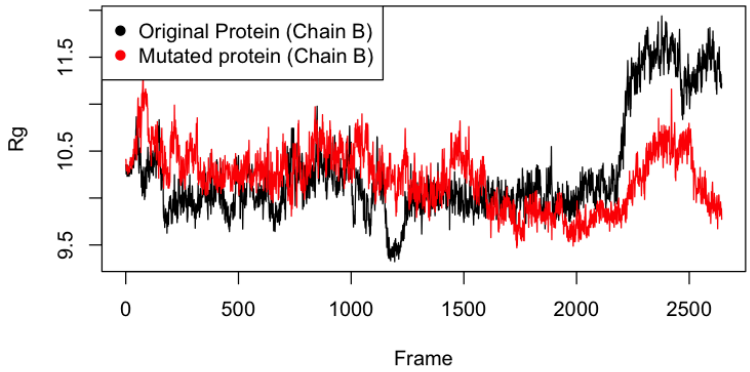
As seen in these two pictures, the majority of the protein was in red, representing a small RMSF value. This was because the scale was skewed by the residues at the ends with high RMSF values shown in white and blue in the picture. These two pictures prove the chain B graph above true, as the control residue had a higher RMSF value for the first few residues, while the mutated residue had a higher RMSF value for the last few residues.

Radius of gyration

I measured the radius of gyration for the two proteins, as shown below:

Chain A:





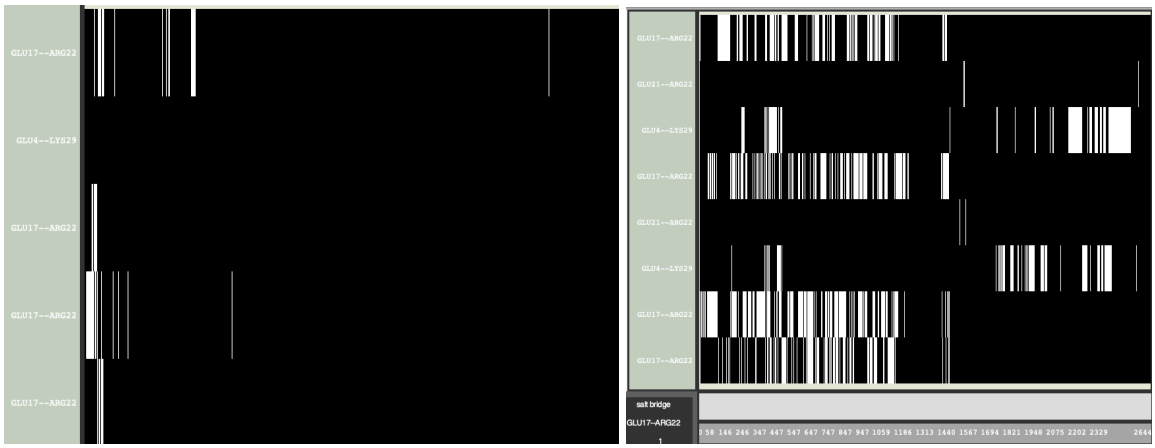
The radii of gyration trends were similar to those of the RMSD trends. For Chain A, there was an increase followed by a decrease between frame 250 and frame 750 on the control protein but not on the mutated protein. For chain B, the control protein experienced a large increase after frame 2100, while the mutated protein didn't experience an increase that was as large.

Salt Bridge Timeline

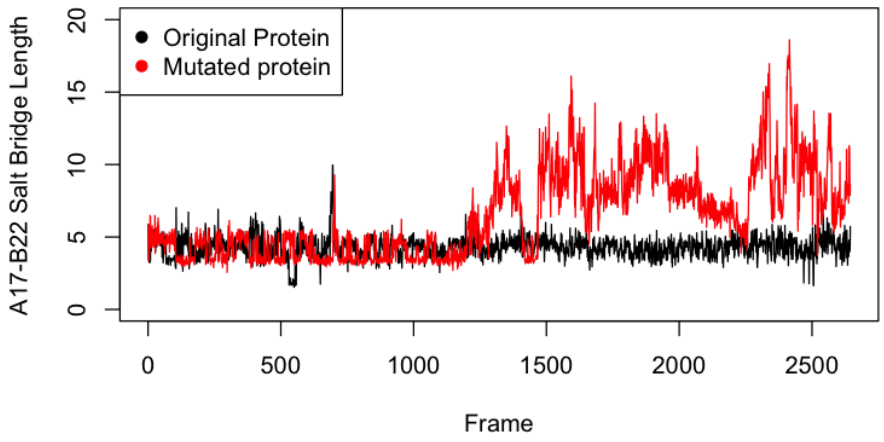
I used VMD's timeline tool to create a timeline of the salt bridges formed and broken within the proteins. I set a cutoff length to 3.2 angstroms.

Control (3.2 angstroms):

Mutation (3.2 angstroms):



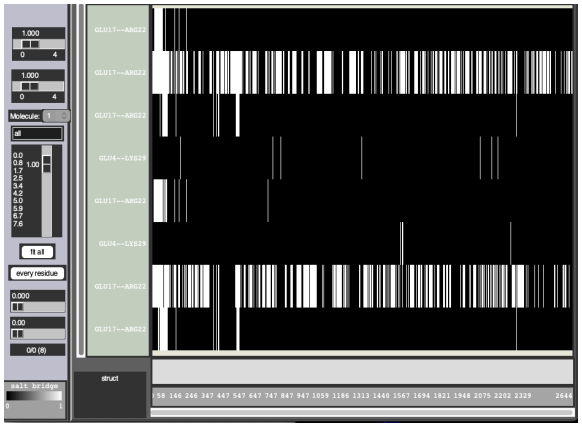
As seen in these two charts, the mutated protein had salt bridges that lasted significantly longer. The salt bridges on the control protein started off with very little appearance, and almost completely disappeared. In comparison, there were consistently salt bridges on the mutated protein throughout the first half of the simulation, and new salt bridges appeared in the second half. In order to investigate what the specific situation was, I plotted the distance between the two ions for the A17-B22 salt bridge, one of the salt bridges that appeared throughout both proteins.



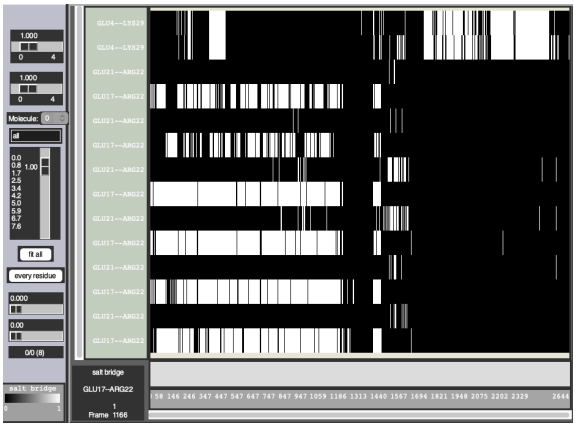
This graph suggests the opposite of what the timeline suggested. Whereas in the timeline the mutated protein had more salt bridge activity than the control protein, this graph shows that the A17-B22 salt bridge was less prominent on the mutated protein. The graph shows that the control protein had a very consistent distance the whole time. For the mutated protein, the distance was very stable in the beginning, yet rose from about 5 angstroms to more than 10 angstroms at certain points and never lowered back down, suggesting that the bridge was broken.

A possible cause of this was that I set my cutoff too low. The distance between the two ions was about 5 angstroms on average, higher than the cutoff of 3.2 angstroms. The reason the mutated protein was shown to the salt bridge for longer was because in the first half, the length dipped to less than 3.2 angstroms more frequently than the control protein did. To see what it was like with a higher cutoff, I drew the following two timeline plots using a cutoff of 5.2 angstroms.

Control (5.2 angstroms):



Mutation (5.2 angstroms):

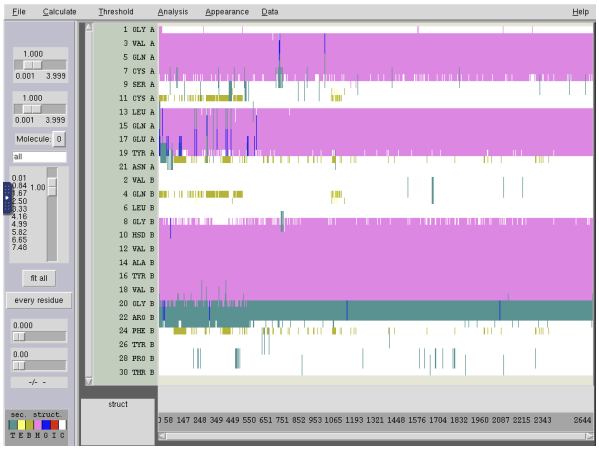


This fits with the graph. In the control protein, the salt bridge lasted throughout the entire simulation, while the salt bridge for the mutation protein only lasted for half of the simulation before it virtually disappeared.

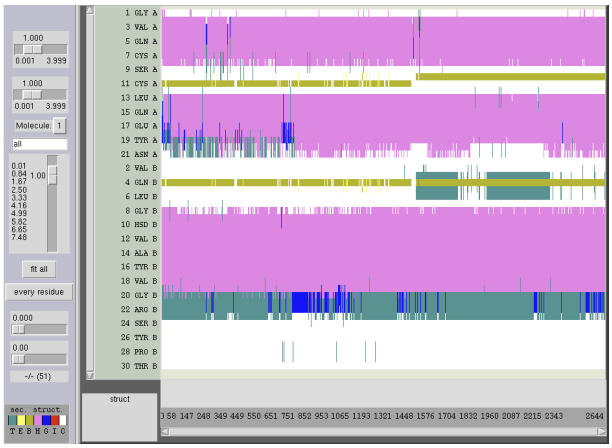
Secondary Structure Timeline

I used the VMD timeline tool to plot out the history of the secondary structure of the control protein and the mutation.

Control protein



Mutated protein



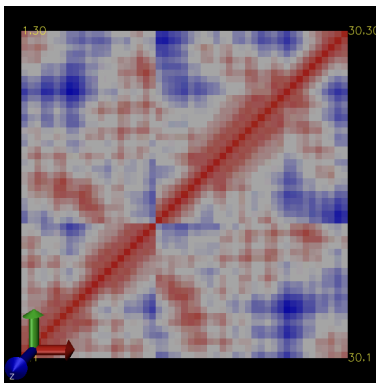
There were no major changes to the structures. However, there were a few small changes:

- The alpha helix on the mutated protein extended to residues A19 to A21 several times on the mutated protein. The same didn't happen to the control protein.
- There were some isolated bridges (yellow) in the beginning in places such as CYS A11, CYS A20, and PHE B24 in the control protein that gradually diminished over time. The bridges in CYSA11 and CYSA20 lasted much longer in the mutated protein, almost spanning the entire simulation. However, the B24 bridge in the mutated protein didn't exist as that was the position in which the mutation happened.
- The number of molecules in turns (turquoise) decreased slightly for the control protein. GLY B23 was originally part of a turn but stopped being part of it after about frame 600. For the mutation protein, the number of molecules in turns stayed the same. However, that turn turned into a 3-10 helix (dark blue) a few times during the simulation.
- A turn between residue B3 and residue B6 appeared in the middle of the simulation on the mutated protein. It disappeared towards the end of the simulation.

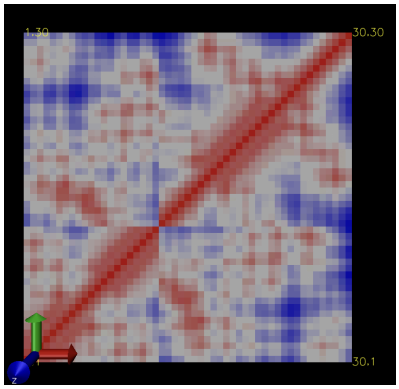
Distances between molecules

I used a function to draw a chart mapping the distances between each residue. The following were my results for the control protein:

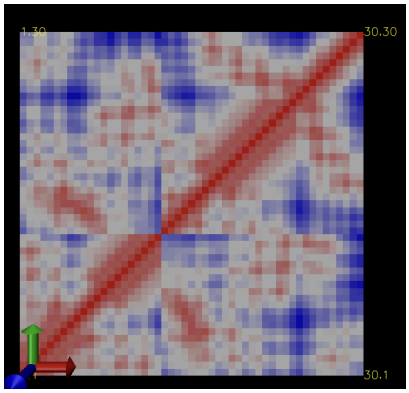
Frame 500:



Frame 1500:



Frame 2500:



In this chart, red represents a distance of 0. Since the neighboring amino acids are the closest to each other, the red line that goes diagonally through the center makes sense. It also makes sense that the part 2/5 of the way up isn't red, because the protein is composed of 2 chains with one being 21 residues long and one being 30 residues.

The plot almost didn't change. A minor difference, however, was that the bottom-left area became bluer than it originally was. This means that parts of chain B went further from parts of chain A.

Discussion

RMSD, RMSF, and Rg Analysis

The biggest findings from the RMSD, RMSF, and Rg analysis included the “stretching” event on chain A of the control protein close to the beginning and the “unfolding” event that happened to chain B of the control protein towards the end of the simulation. Both of these events didn’t occur in the mutated protein. They resulted in the Rg and RMSD values of the control protein rising by a lot compared with the mutated protein. In addition, the “unfolding” event likely caused the RMSF values on the ends of chain B of the control protein. The lack of the “stretching” and “unfolding” in the mutated protein could have been the cause for the insulin lacking biological activity under the PheB24Ser mutation, as discussed in the papers in the introduction. It is possible that the “stretching” and “unfolding” are necessary steps in binding with the insulin receptor proteins.

Salt Bridge Analysis

When I first looked at the timelines of the two proteins (under a 3.2 angstrom cutoff distance), I thought that the presence of the salt bridge in the control protein was less significant than that in the mutated protein. However, when I plotted the bridge distances out over time, I discovered that the mutated protein might have been the one with less salt bridge presence and that I turned the cutoff distance too low. Hence, I drew another timeline with a cutoff of 5.2 angstroms, and saw that under this constraint, the salt bridge existed in the control protein the entire time but only halfway through the simulation in the mutated protein.

Of course, there is the possibility that the salt bridge was only formed when the length was less than 3.2, and that the control protein almost never formed the salt bridge in the first place. Hence, this would require further investigation.

The salt bridge being studied was the salt bridge between position A17 and position B22. A paper showed that this salt bridge exists, and is related to another salt bridge consisting of glutamine at position A21 and the arginine at position B22 (Le Nguyen, 2025). This paper also said that the arginine in the position B22 is very significant for the activity of insulin as it provides a salt bridge and plays an important part in the folding. It suggests that the reason that guinea pig insulin has low activity is because it doesn’t have arginine at B22 (Rose, 1981).

If my prediction that the A17-B22 salt bridge was more prominent in the control protein was true, then the results of these two papers would be proven correct. Papers mentioned previously suggest that the control protein has more biological activity than the mutation protein. If the salt bridge is crucial for the biological activity of the protein, then it explains why it was much more significant in the control protein than in the mutation protein.

Secondary Structure analysis

The secondary structure of the mutated protein was not completely different from the secondary structure for the control protein, but there were some minor changes. For instance, the mutated protein had instances of isolated bridge structures that lasted for the entire simulation, unlike on the control protein where the bridges disappeared less than halfway into the simulation. Also, there were some 3-10 helices and turns that formed in places where they didn't form in the control protein. These changes may or may not be related to the lack of activity in the mutated protein that several of the papers touched on. However, it is hard to determine which change caused it.

One of the papers suggested that the B24 mutation would cause the protein to have less alpha helix content. However, this proved to be the opposite in the simulation. Not only did the mutated protein have all of the alpha helices of the control protein, but at times, the mutated protein had one of its alpha helices extend further, which never happened on the control protein. Hence, this needs to be further investigated.

Insights provided by MD Simulations

The MD simulations provided a lot of information that could not have been deduced from the crystal structure. At the beginning of the simulation, the control protein and the mutation protein nearly looked identical, and there was not much to investigate as there was only one amino acid being changed. However, as I ran the simulations, I discovered many different things from the moving molecules, such as the "unfolding" movement of one of the ends of chain B of the control protein rotating and changing its orientation. In addition, I also discovered several changes to the salt bridges and secondary structure over time, which I wouldn't have discovered if I only looked at the crystal structure. There were several differences in the secondary structure of the mutation protein compared with the control protein, such as 3-10 helices and turns appearing in different places. These could have been the cause of the lack of activity in the mutated insulin described by the papers in the introduction. Also, one salt bridge was much more prominent in the control protein than in the mutation protein, which may be the explanation for why the mutation protein had less activity, according to the papers I read.

Pitfalls of the MD simulations

However, there are also pitfalls with MD simulations. Throughout this experiment, I only modelled the movement of a single insulin molecule in water. In a real biological environment, the insulin will be moving around and interacting with other things such as other proteins or insulin molecules, which my simulation didn't simulate. In addition, the binding of insulin to insulin receptors is a key part of the function of insulin which I didn't look at in my simulation.

In addition, there are limitations to MD simulations that make them not completely realistic. For instance, I used TIP3P water in my simulation to reduce computer resource usage, which isn't very accurate. This may hinder the accuracy of the simulation.

Further Analyses

Like mentioned above, the simulation was done in a small water box environment with only the insulin protein itself. It is completely different from a realistic environment which would have cells and proteins floating around. Furthermore, the main function of insulin is to bind with the insulin receptors, which weren't even included in the simulation. Hence, a more realistic simulation of insulin should be much, much bigger. It should include a realistic human body environment with blood cells, other proteins, and in particular, insulin receptors. If there were sufficient computing power, the best simulation would be that of an entire human as it would be completely authentic, or even planet Earth or the universe.

Another change would be to improve the models of atoms and water. Many approximations were done in the simulation to reduce computing cost, such as using TIP3P, an inaccurate model of water, to represent water and using springs to model bonds. To increase the amount of realism, more realistic and computing power-intensive models should be used. Also, the simulation only simulated the motion of particles, and did not take into account the various chemical reactions or electron movements in the molecules. Thus, the quantum-mechanical forces that cause things such as chemical bonding/unbonding should be included in the simulation.

Finally, the time scale should be increased if there is enough computing power. My simulation of 50 nanoseconds nanoseconds managed to provide some insight into the motion of insulin. However, the "unfolding" action of one end of chain B unbending that I found interesting happened near the end of the simulation and I wasn't able to see what happened after the event. Hence, increasing the length of my simulation would allow me to know a lot more about insulin. I will be able to know if this action is something that periodically happens, and whether or not it can also happen to the mutated protein.

Conclusion

To conclude, MD simulations proved to be a very useful tool for analyzing proteins. In this report, I managed to find many changes caused by the mutation of PheB24Ser in human insulin. Most of my findings were consistent with the information in the papers I looked at, meaning that my simulation was likely correct. However, there were parts that didn't agree, such as when the alpha helix content increased under the mutated protein when the papers predicted that the content would decrease. Hence, this would require more detailed simulations.

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